7. Facing Bets - End of Action Spots

An **End of Action Spot** is a situation in which Hero faces a bet or a raise and is is considering either folding or calling, and if he calls, that will be the last action taken in the hand by any player. Such spots occur when facing an all-in on any street or a bet or raise on the river.

Until now we've been the post-flop aggressor. Let's switch the roles around and see what happens in a very common and excruciatingly annoying spot to be in - when we're facing a bet or raise that will be the last bet or raise of the hand and don't know whether to call or fold.

7.1 The Two-Part Thought Process

Let's jump straight into a hand. The third (green) number on the HUD in Hand 48 below is **Raised First In** from the relevant position - in this case, the HJ.

Hand 48

UTG – Unknown
HJ – Reg (25/20/19)
CO – Unknown
BU – Hero
SB – Unknown
BB – Unknown

Dealt to Hero:

UTG folds, **HJ raises to 3BB**, CO folds, **Hero calls 3BB**, SB folds, BB folds.

Flop (7.5BB, 2 players)

HJ bets 4.5BB, **Hero calls 4.5BB**.

Turn (16.5BB, 2 players)

HJ bets 12BB, **Hero calls 12BB**.

River (40.5BB, 2 players)

SB bets 26BB, **Hero?**

This is the type of spot people post on my student forum more than anything else. It's one of the most

uncomfortable spots to face in poker. Hero faces a river bet and his decision will be to either call or fold. It's not that turning his hand into a bluff by raising is necessarily out of the question, but we'll leave this more advanced kind of play for now.

Assuming that Hero is choosing between calling and folding, this spot is known as an **end of action spot**.

These are some of the most common spots in which improving players continuously make big mistakes in their thought process. Let me give you an example of how not to approach this spot. See if you can spot a few problems in the following mental dialogue concerning the river decision in this hand.

"Well he's bet really big so he might think that I think he's bluffing and trying to get me to call, or is he trying to bluff me off my hand because he thinks I have something like I have? He would probably bet smaller with a value hand actually so I should call because he's trying to push me off the pot."

This thought process is a train wreck of unjustified conclusions. Hero is irrationally trying to soul read the very thoughts of some person across the world somewhere sat in front of a computer screen that he knows almost nothing about other than that the player is a Reg. Hero could leap with equal ease to either conclusion that Villain is more likely or less likely to be bluffing with this sizing using his amateur clairvoyance. The truth is that he simply doesn't know what the sizing means yet, if it means anything at all.

Perhaps the biggest problem with Hero's thought process, however, is that he hasn't even considered how much equity he needs (how often he needs the best hand) to call here. This is the absolute staple of the situation and should be the very first thing on Hero's mind. The rest of his decision-making should go out from there.

Only when Hero has at least roughly gauged this required equity should he go about estimating if he actually has that much equity on average by thinking about how likely Villain is to be bluffing vs. value betting.

The correct thought process can thus be clearly broken into two parts:

Part 1: What is my required equity to call?

Part 2: Do I have that much equity on average?

Part 1: Required Equity

We need to first calculate our equity target before we can decide if we should expect to win that often on average.

Required Equity – End of Action Spots

We call the amount of equity needed to break even on calling a bet in an end of action spot **RE (Required Equity)**

We call the bet size we're facing, but have not yet invested **ATC (Amount To Call)**

We call the total pot size **including** Villain's bet **TP (Total Pot)**

Now we've defined our variables we can state the following mathematical rule. This is the first step to working out whether or not to call a bet in an end of action spot.

$$RE = \frac{ATC}{ATC + TP}$$

Example: Hero faces a 13BB bet into a pot of 20BB on the river and is deciding whether to call or fold.

ATC = 13. Hero would have to invest 13BB to call.
TP = 33. This is the pot after Villain has bet.

RE = ATC / (ATC + TP)
RE = 13 / (13 + 33)
RE = **28.2%**

Pitfall: The most common mathematical mistake made in poker is to divide 13 into 33 instead of into (13 + 33). If we call and win, we get our 13BB back as well as Villain's 13BB and the pot as it was before he bet. We thus need to add our 13BB to the amount we're calling to win. We are not inserting those 13BB into a vending machine to have a chance of winning a separate pot, they will come back to us when we have the best hand.

Let's do this calculation for Hand 48.

Hero is facing a 26BB bet into a pot of 40.5BB

ATC = 26

TP = 66.5BB (again including Villain's bet, do not forget to add this to the pot)

RE = ATC / (ATC + TP)

RE = 26 / (26 + 66.5)

RE = 28.1%

That's step one of our thought process complete, and that's all very well for out of game analysis, but how is Hero supposed to do that calculation so precisely with the limited amount of time available in game?

He's not. Instead he should refer to milestone amounts of required equity corresponding to the rough size of Villain's bet in relation to the pot. If he knows how much equity he needs to call a pot-sized bet, a two thirds pot sized bet, a half pot-sized bet and so on, he can quickly estimate the rough amount of equity he needs by looking at the nearest milestone to the bet-size in question.

Bet Size Faced	Required Equity
Bet into Empty Pot	50%
Pot Sized Bet	33%
¾ Pot	30%
⅔ Pot	29%
½ Pot	25%
⅓ Pot	20%
¼ Pot	17%
⅕ Pot	14%

Figure 37 - RE Milestones

I strongly recommend that you take the time to ingrain these numbers into your subconscious so that they can effortlessly pop up to the forefront of your thought process when you face a bet in an end of

action spot. This way you'll save the rest of your time bank for figuring out part two of the puzzle, and as you might have guessed, that's the tough bit.

In Hand 48, Hero looks at Villains bet size and sees that it's a bit more than half pot. He reasons that he therefore needs a bit more than 25% equity to call the bet.

Part 2 - Actual Equity

Are we good just over a quarter of the time in Hand 48?

Sometimes my students ask me how to mathematically calculate their equity in an end of action spot. There is no way to magically conjure up the amount of equity we have in the hand or even the amount we have against our opponent's range. Instead Hero must go into the tank and consider the following factors:

- **Q1. Is it likely that Villain can be value betting many worse hands?**

  If YES, we can usually make a quick call.

- **Q2. Does Villain get to the river with any air in his range?**

  If NO, we can usually make a quick fold.

- **Q3. Does Villain credibly represent better hands?**

  If NO, we can usually make a bluff catching quick call.

- **Q4. What else do my population/player type/Villain reads tell me?**

  This is more complicated and something that requires solid analysis.

Question 1: Value Betting Worse

One of the surest indicators that calling must at least be a reasonable option if not a fantastic one is when Hero's hand is so strong that it actually beats some of Villain's value range. In these cases Hero needs Villain to be bluffing far less often in order to call as he can beat hands that Villain isn't even bluffing with.

In Hand 48, it doesn't seem like Villain should be value betting anything worse. To blast three streets with **88** or **99** for value is incredibly optimistic if not delusional and so with Villain being a Reg with some presumed degree of competence, we can rule that out.

Answering no to the first question doesn't in and of itself mean that we should fold; it only means that we don't have an easy call.

Question 2: Can Villain Have Air?

If Villain's range contains no air, then the likelihood that he is bluffing decreases significantly. In some situations in which all of the hands that Villain gets to the river with have either improved or have decent SDV, Hero will be able to find folds more easily. As we saw back in Chapter 5 Hand 41, when Hero c-bets flop and turn and then a draw completes on the river, check/calling the river is a horrible option for Hero to take with a marginal hand. This is because Villain's range is probably airless, meaning he either has showdown that he's likely to check, or a stronger hand like a set or flush to bet for value with if Hero checks.

In Hand 48, Villain could be bluffing three streets with some air hand such as **AK** or **AJ** - these hands picked up equity on the turn and so make some sense. Much of Villain's air, though, mainly that containing a **Q**, has now improved to beat us.

Just because it's possible that Villain can have air in his range does not mean that he'll choose to bluff all of that air on the river and doesn't mean we can just click call. It's just another preliminary factor to look at. If Villain couldn't have air, however, a fold would be an easier choice.

Question 3: Representing Better Hands

Some weaker players bluff in spots where their line just doesn't make sense for value. For example, if an mindlessly aggressive player checks back the turn capping his range to only weak top pairs and worse and then raises a river bet, Hero can feel a lot better about calling that raise lightly, provided that he has a solid enough read that Villain always bets his strong hands on the turn. Being able to detect when a Villain's line doesn't make any sense for value is a skill that comes with practice. A word of caution though, Passive Fish will slowplay big hands very often so one of these players checking flop and turn then shoving his stack in on the river is not the same as the spot vs. the aggressive player with his capped range trying to represent the nuts.

In Hand 48, Villain's range is very uncapped and he credibly represents lots of value hands ranging from **QJ** and **KQ** to overpairs and sets.

This doesn't mean that Hero should fold, but if Villain's line had been dubious for value, which it is not, then calling would very likely have been good.

Question 4: Using Reads

Three questions down and we're no closer to an answer. Nevertheless, those three questions are important to consider as they can sometimes make the process much easier and lead us to a faster answer. Sadly, most of the time that we face an end of action spots on the river, we won't be able to answer questions one to three in a way that solves the problem straight away. Villain's line will normally make sense for value, ruling out air won't normally be possible, and his betting range won't normally contain worse value hands than ours.

In Hand 48 we have the dreaded bluff catcher in a spot where there's no obvious reason to either fold

or call.

So let's look at our reads. There are three specificities of read ranging from most useful to least useful from left to right in Figure 38 below. Nevertheless, any read is better than none at all.

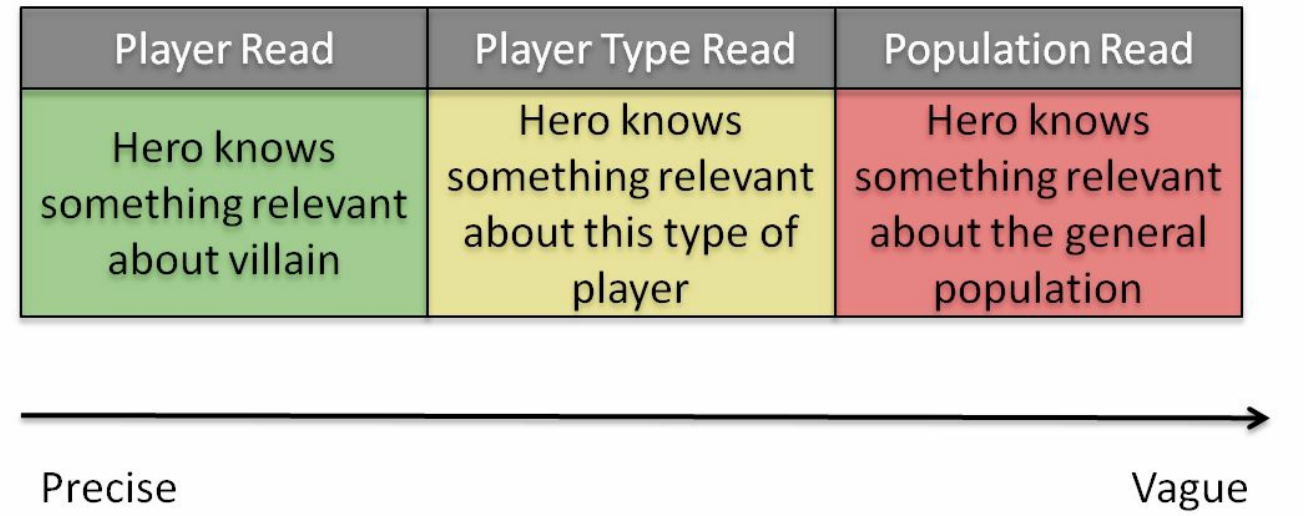


Figure 38 - The Three Specificities of Read

Player Reads concern things that Hero has observed either from that specific Villain's past actions, his showdowns or from his HUD stats. Any notes Hero has taken on Villain also fall into this category. This is the most useful type of read and gives the most accurate information to sway Hero's decision.

Player Type Reads concern things that are true of the category of players to which Villain belongs. Regs tend not to slowplay the nuts for no reason for two streets. Fish tend to call with a wider range than they bet. Aggro Fish tend to have a wide array of bluffs in their range and might even turn SDV into bluffs for no good reason. The list goes on...

Population Reads concern things that Hero knows about the general Reg population. Perhaps the population folds too much to c-bets. Maybe it tends to have unbalanced 4-bet ranges towards bluffs. Perhaps we're at 5NL and the population has trouble finding the fold button.

It's Hero's job to decide which level of specificity he'll be using, but he never has the excuse: '***I know nothing***.' At the very least he should have a cache of population reads in his head. I advise you to study the average population in your games and develop a feel for the most reliable population tendencies among the Regs there and think about how to exploit these tendencies.

In this hand we know nothing other than that Villain is a Reg. This is not a useful player type read. If Villain were a crazy Aggro Fish, then we'd have a very useful one indeed. Hero should refer to population reads and ask himself how often the average Reg in his games fires this board for three streets as a bluff.

The answer in most low stakes games will be 'not very much', or at least 'not enough' and so Hero should struggle, unless he plays in a very aggressive game, to find the requisite 28% equity to call with **TT**. I'd imagine in most low and microstakes games that the Reg population typically has a

predominantly value orientated range when they bet three times like this.

One final point is that Hero must remember that he, like Villain, has a range of hands that he gets to the river with and that range contains a good few stronger hands like **AQ** and sets. Folding this one hand doesn't make Villain's play great overall if he ***is*** bluffing. Hero is not making a mistake if Villain rolls over a bluff after Hero folds in this one instance - Villain can only exploit Hero if Hero folds too often, not if he folds a hand that is at the bottom of his range.

We'll see a lot more about this approach of playing our whole range rather than just our hand as the manual goes on. Chapter 13 will deal with spots from this perspective.

In summary, we used a two-part thought process to come to our decision in Hand 48. We first found our required equity (RE) and only then asked three questions to see if we could solve the spot quickly. When that failed, we assessed our population reads as none of the more specific reads could help us and decided that we'd probably lose here more than 72% of the time on average if we called the bet.

Hero Folds.

Out of game Hero can be more specific than this and input an estimated range of hands for Villain into an equity calculating poker program. This will give him his exact equity if Villain's range was exactly the model Hero used in the calculation. Of course Hero can't soul read Villain's range out of game any more than he can in game, but such a tool can help draw conclusions such as: "If Villain bluffs all of his **AK** and **AJ**, and then the rest of his range is value, then I should call."

In fact vs a range of **[AJ, AK]** as bluffs and then **[AQ KK AA A4s 22 33 55 99 QQ]** for value Hero can use such a program to calculate that his equity is 43% and that he can call comfortably. Due to our population reads, however, we decided that Villain probably wasn't bluffing anywhere near this much on average and decided to fold for that reason.

Here is a flowchart to serve as a summary of how we solved this end of action spot.

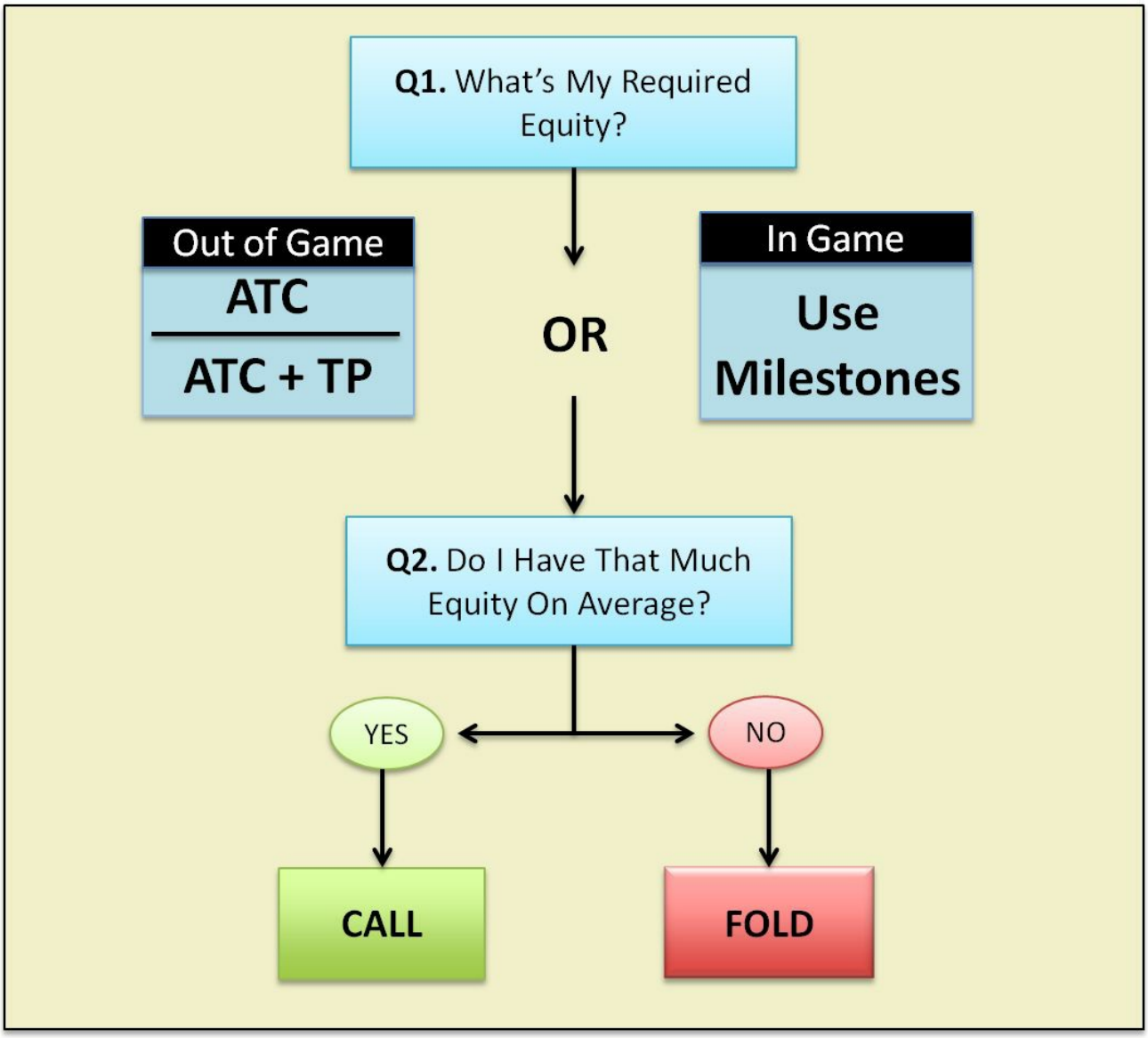


Figure 39 - A Thought Process for End of Action Spots.

7.2 Stats and Examples

Quite often Hero will be much less clueless than he was in Hand 48 and will have useful HUD stats to hand. Let's have a look at some of the best stats for deciphering how aggressive Villains are with the betting lead and deducing whether we should call in an end of action spot.

CBet Flop

Definition: CBet Flop shows the percentage of the time that a player bets the flop as the pre-flop aggressor given that he had the opportunity to do so. Higher values show that when Villain bets the flop, he will on average reach the turn with a larger amount of his overall flop range.

Reliability (No. of Hands)
50 = **Poor:** Likely only over 5-10 opportunities with the average active Reg.
100 = **Okay**: Gives a rough flavour with players who raise a decent number of hands pre-flop. Players who raise rarely will have had less opportunities to c-bet.
300 = **Decent**
500 = Good
1000 = Excellent

Interpretation
Caution: This stat does not accurately represent the percentage of the time a player c-bets the flop in the spot at hand, but shows an average c-bet% for all flops. Some students take this stat too literally. Just because a competent Villain c-bets 70% does not mean that he c-bets as much as 70% of his range on a very wet flop five-handed. Similarly, on the driest flop heads up, he probably c-bets more than 70%.

0-50 = Very Low: This player plays very straightforwardly and mainly c-bets for value.
50-60 = Low: This player does make light c-bets but not as much as he should.
60-65 = Average: This player likely selects his light c-bets well.
65-75 = High: This player likely makes some light c-bets that he shouldn't.
76+ = Very High: This player c-bets far too much.

Although the flop will not usually be an end of action spot, using **CBet Flop** is very important for deciding how lightly to call river bets too where Villain is firing three streets at us. The larger the quantity of light c-bets he makes on the flop, the more air he'll take to the turn. This will often

translate to him having more air to bluff with on the river as well, but before making our decision we need to look at how often he fires those later streets.

CBet Turn

Definition: CBet Turn shows the percentage of the time a player bets the turn as the pre-flop aggressor given that he c-bet the flop and had the opportunity to do so. Higher values show that Villain will on average reach the river with a wider range after he bets turn.

Reliability (No. of Hands)
Note that players who c-bet the flop a lot will have more opportunities to c-bet the turn.

50 = **Awful:** Likely only over one or two opportunities.
100 = **Poor**
500 = Decent
1000 = Good
5000 = Excellent

Interpretation
Caution: Again, this stat cannot be taken literally in all situations. If a competent player c-bets the turn in a multi-way pot on an anonymous card, it's more likely that he has mainly value hands. If he bets the turn heads up on a scare card, it's likely that he has more air.

Interpreting this stat also depends massively on Villain's **CBet Flop** stat. The wider the range he takes to the turn, the more indicative a high value in this stat will be of bluffs. The tighter his flop c-bet range, the more value he'll have on the turn even if this stat shows a high value.

0-35 = Very Low
35-50 = Low
50-60 = Average
60-70 = High
70+ = Very High

But I stress this again, it depends so much on how wide Villain c-bets the flop.

So it's prudent to use the last two stats together when considering turn decisions and not just this one

in isolation. The same goes for the river where we are most likely to encounter the end of action spots that examined in this chapter.

CBet River

Definition: Cbet River shows the percentage of the time a player bets the river as the pre-flop aggressor given that he c-bet the flop and turn and had the opportunity to do so. Higher values show that Villain is more likely to bet the range he reaches the river with.

Reliability (No. of Hands)
Players who c-bet the flop and turn a lot will have more opportunities to c-bet the river. This stats takes even longer to converge than **Cbet Turn**.

100 = **Awful**
500 = **Poor**
1000 = Decent
5000 = Good
10000 = Excellent

Interpretation
Caution: Again, this stat cannot be taken literally in all situations. Some rivers complete draws and reduce the amount of air a player can have, some river cards will rarely get bluffed while others will more frequently.

Again, the higher the player's **Cbet Flop** and **CBet Turn** stats, the more air he'll have to use as a bluff on the river. A player who rarely gets to the river with weak hands will naturally have a high value in this stat; it doesn't make his range bluff-heavy.

0-35 = Very Low
35-50 = Low
50-65 = Average
65-70 = High
70+ = Very High

The ratio of these three stats over a good sample can shed some light on Villain's river bluffing frequencies. It's very important that Hero assesses them in relation to each other and not in isolation.

Before we look at a few hands. There is one more general stat that can be useful over smaller samples to give us an idea of how often Villain wins the pot when he sees the flop and indicates how much he fights for pots overall and tries to generate fold equity with the non-value parts of his range.

WWSF

Definition: WWSF stands for **Won When Saw Flop** and shows the percentage of the time a player won the pot having seen the flop. It is a very good general indicator of aggression and willingness to try to make opponents fold in all post-flop situations.

Reliability (No of Hands)
100 = **Poor**
500 = **Decent**
1000 = **Good**
3000 = Excellent
5000 = Accurate

Interpretation
0-37 = Very Low: This player is extremely weak/tight post-flop and never fights for pots without a strong hand.
37-42 = Low: This player gives up too much post-flop.
42-47 = Average (low): This player is on the straightforward side of average.
47-52 = Average (high): This player fights for more pots than the average Reg.
52-55 = High: This player likely bluffs too much and refuses to give up in spots where he should.
55+ = Very high: This player is very aggressive and likely unbalanced to bluffs in most situations.

Let's move onto some examples now and put these stats and the previously taught thought process into use.

In the following hands the third (green) number on the HUD is **Raised First In** from the relevant position. The fourth (blue) number is **WWSF** and the next three (brown) numbers are **CBet Flop,**

CBet Turn and **CBet River** respectively.

Hand 49

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Unknown
SB – Reg (27/22/65/51/74/62/65) (300 hands)
BB – Hero

Dealt to Hero:

UTG folds, HJ folds, CO folds, BU folds, **SB raises to 3BB**, **Hero calls 2BB**.

Flop (6BB, 2 players)

SB bets 4BB, **Hero calls 4BB**.

Turn (14BB, 2 players)

SB bets 9BB, **Hero calls 9BB**.

River (32BB, 2 players)

SB bets 29BB, **Hero?**

The first three streets are very standard.

Pre-flop: Hero defends a hand that is in absolutely fine shape vs. a 65% SB RFI in position, closing the action. No problems here.

Flop: Hero calls a c-bet with an extremely strong hand relative to Villain's c-bet range, which as we know from the high 70% stat, includes a lot of worse hands. Hero could exploitatively raise with the right dynamic where Villain is likely to lash out by 3-bet bluffing the flop or perhaps call the raise far too lightly. Without this read, getting 3-bet on the flop is awkward and raising unnecessarily narrows Villain's very wide range. Hero also gets to protect the weaker parts of his range by having hands as strong as this in his flop calling range. This prevents our aggressive opponent from bluffing like crazy and getting enough folds from us overall. We'll look at these kind of decisions more closely in the next chapter.

Turn: The **Kc** is far from the best card but it is one that Villain is very likely to continue betting on with a large part of his range. Hero still has the best hand very often here and so folding is out of the question especially vs. someone who c-bets the turn 62% of the time.

River: And here is the end of action spot. Hero is clearly not raising this river with a hand that is basically a bluff catcher. He's calling or folding and either way his action will be the last one in the hand. Time to initiate the two-part process.

Part 1: Required Equity

Hero needs ATC / (ATC + TP) equity to call the bet. Remember TP includes Villain's bet.

29 / (29 + 61) = **32%**

If we were in game using our milestones, we'd simply reason that Villain's bet is almost pot sized so Hero needs almost 33% equity to call.

Part 2 - Actual Equity

Do we have enough equity against Villain's range to call? Let's go through questions 1-3 before we examine our reads and see if we have an obvious decision available.

Q1. Is it likely that Villain can be value betting worse hands?

This is unlikely, but possible. Villain could potentially be value betting **Q9** or **J9** for thin value, but his large sizing makes this less likely. Even if he is betting these hands, they aren't a large part of his range by any means. No quick call available for this reason.

Q2. Does Villain get to the river with any air in his range?

Absolutely. Lots of draws have missed and we know this player to be capable of getting to the river

with a wide range in this way, especially blind vs. blind where ranges are even wider than usual. No quick fold available here.

Q3. Does Villain credibly represent better hands?

Yes he can have a whole host of **[TT-QQ Kx AA]** and sets. No quick call available for this reason.

Q4. What do my population/player type/player reads tell me?

This player fires post-flop bets frequently on all streets. His **WWSF** is high and he clearly isn't one to play fit-or-fold on the later streets. The **Kc** on the turn is a perceived scare card on which many Villains will bluff a wide range and Hero shouldn't have too much **Kx** in his own range after he calls the flop c-bet. This is a card that is much better for Villain's range than Hero's. All of this makes Villain fairly likely to be bluffing this river with a wide range of hands. Hero is fairly high up in his own range with **A9** and holding an **A** makes some of Villain's value combinations such as **AK** and **AA** less likely.

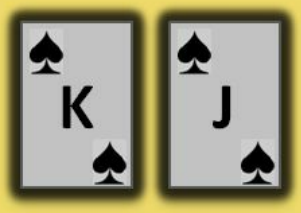
Moreover, the large river sizing seems to discount some thinner value bets. It seems less likely that Villain will show up with **TT**, **JJ**, **QQ**, or a weak **Kx** with this huge sizing as with these hands he needs to be called by very weak holdings. This reduces Villain's value range and makes bluffs a larger proportion of his range overall.

In summary, Hero should comfortably reach his 32% equity target vs Villain's range.

Hero calls 29BB.

Hand 50

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Reg (24/20/56/48/65/51/53) (2200 hands)
SB – Hero
BB – Passive Fish (56/12/36/40/40/25/0) (90 hands)

Dealt to Hero: 

UTG folds, HJ folds, CO folds, **BU raises to 3BB**, **Hero calls 2.5BB**, BB folds.

Flop (7BB, 2 players)

Hero checks, **BU bets 5BB**, **Hero calls 5BB**.

Turn (17BB, 2 players)

Hero checks, **BU bets 10BB**, **Hero calls 10BB**.

River (37BB, 2 players)

Hero checks, **BU bets 20BB**, **Hero?**

Let's assess this hand street by street, Even though the earlier streets are not directly relevant to the

end of action thought process, the concepts there are both good revision and good warm ups for what's to come in later chapters.

Pre-flop: Hero wouldn't normally have a flatting range out of the SB as we discussed here in Section 6.3. In this spot, however, the Fish in the BB provides an exception to the rule and Hero happily flats **KJs** since the Fish will so often be in the pot making all of his horrible mistakes. **KJs** is of course also in fine shape vs. the opener's wide stealing range from the BU.

Flop: Sadly, the Fish did not oblige and folded pre-flop. Hero now calls a c-bet in a heads-up pot with a flush draw and two overcards. Raising is another +EV option but Hero reasons that his hand is in such good shape and he can call on any turn card, so elects to use his weaker draws as check/raise bluffs, keeping this one in the calling range. He also notes that Villain could continue bluffing when he has air on the perceived scare card turns where Hero in fact makes top pair to go with his draw.

Turn: This is exactly what happens. Hero picks up heaps of SDV to go with his flush draw and happily check/calls the turn, expecting to be miles ahead of Villain's betting range.

River: Part one of our thought process tells us that we need just over 25% equity to call this half pot sized bet. Out of game we can calculate this to an exact 26% using **ATC / (ATC +TP)**.

Hero asks **Q1.** Can Villain value bet worse hands? It seems he can in this spot. Given that the turn is a **K** - a scare card on which Villain can be expected to bet a wide range. Hero could very easily feel obliged to call down with hands like **TT**, **98, A9** etc. and should quite rarely have a better hand than these.

We already have an answer. Not only can we beat some value hands, but we only need to be good just over a quarter of the time to make calling better than folding. Villain can also have some air here given Hero's weak range and good perceived fold equity so this makes the call even more obvious.

Hero calls 20BB.

For the first time, our initial three questions have led us straight to an answer without having to scrutinise any reads. This call isn't close.

Hand 51

UTG – Passive Fish (40/14/7/32/55/30/25)
HJ – Unknown
CO – Unknown
BU – Hero
SB – Unknown
BB – Unknown

Dealt to Hero:

UTG raises to 2BB, HJ folds, CO folds, **Hero calls 2BB**, SB folds, BB folds.

Flop (5.5BB, 2 players)

UTG bets 3BB, **Hero calls 3BB**.

Turn (11.5BB, 2 players)

UTG checks, **Hero bets 8BB**, **UTG calls 8BB**.

River (27.5BB, 2 players)

UTG bets 25BB, **Hero?**

Pre-flop: Hero has a pretty easy call here. 3-betting is out of the question vs. such a tight opening

range. Hero has no interest in bloating the pot, and merely wants to flop a set or play favourable board textures in position vs a weaker player.

Flop: Villain c-bets on the small side. This doesn't narrow his pre-flop range much, which on this flop is overpairs, some missed overcards and some flush draws with maybe the odd **77** or **88** depending on whether Villain opens these hands from UTG. Hero calls with SDV against this range, his hand not being strong enough to raise for value and turning his hand into a bluff being a horrible option vs. a player who simply won't fold better hands.

Turn: Villain's check doesn't necessarily narrow his range too much. Passive Fish will be irrationally terrified of a flush here and will automatically check overpairs as well as air on this card very often. Hero bets for value against some flush draws and possibly medium pairs and protects his equity vs the overcards Villain is check/folding now. Villain's call indicates that he is unlikely to have naked overcards with no flush draw. Overpairs, underpairs and overcards with a good club are certainly in his range. It's also not out of the question for Villain to be going into slowplay mode with the nuts (AcKc for example) as passive Fish like to do.

River: Hero makes a 9-high flush and Villain suddenly wakes up and blasts a near pot-sized bet. Hero thinks.

Hero knows that he needs very close to 33% equity to call here since Villain has almost potted it.

He goes through his questions:

Q1. Can Villain be value betting worse? Almost certainly not: it is very unlikely that Villain can have lower clubs to begin with given his UTG opening range and even if he does have **7c7x** or **8c8x** the chances of him blasting away with these marginal hands being such a passive player are very low.

Q2. Can Villain have air? Here we get to answer no and reach an easy decision. If Hero has a bluff catcher and Villain is very unlikely to be bluffing, then Hero needs to fold, but why is Villain so unlikely to be bluffing?

Recall that Villain called the turn bet with two main types of hands, those which are overpairs and have SDV, with or without a club and those which were naked club draws with **AcKx** and the like. The latter bunch are now the nuts and Hero loses to them. The former, when they don't have a flush, are very unlikely to just turn themselves into bluffs given their SDV and how passive Villain is in general. Hero has close to 0% equity vs Villain's range and his decision is now clear.

Hero folds

Hand 52

UTG – Unknown
HJ – Unknown
CO – Aggro Reg (31/25/32/**51**/82/65/70) (425 hands)
BU – Hero
SB – Unknown
BB – Unknown

Dealt to Hero:

UTG folds, HJ folds, **CO raises to 3BB**, **Hero calls 3BB**, SB folds, BB folds.

Flop (7.5BB, 2 players)

CO bets 7BB, **Hero calls 7BB**.

Turn (21.5BB, 2 players)

CO checks, **Hero checks**.

River (21.5BB, 2 players)

CO bets 27BB, **Hero?**

Pre-flop: Hero faces an open from a very aggressive Reg. He reasons that his implied odds are

sufficient here in position vs. an opponent likely to put a lot of money into the pot post-flop. This hand also has the frequent strength to hold its own with one pair on some boards where it doesn't make the straights, flushes, two pair and trips that we'd rather flop.

Flop: Villain has a flop c-bet stat of 82%. This is massively high and Hero knows that his middle pair is comfortably ahead of Villain's wide range, which is still close to the 32% of hands that Villain opens in the CO.

Turn: Villain checks which probably narrows his range considerably to weaker holdings. Hero has no interest in getting check/raised here and heads for showdown with his middle pair. Sure Hero's hand is frequently best, but he can't exactly bet for value and by checking he can possibly induce some bluffs on some river cards.

River: Villain suddenly overbets the pot. Let's go through our thought process.

Hero faces a bet that is a little more than the pot. This means he'll need proportionately more than 33% equity to call here, but nowhere near 50% since Villain is far from betting into an empty pot. He estimates he needs somewhere in the high 30%s to call here. The true target equity found form doing the maths is **36%**.

Q1. Villain value betting worse hands would be absurd here.

Q2. Villain could very easily have air. He's an aggressive player with an extremely wide range. Making a crazy bluff here would not be out of the realms of possibility for this player type.

Q3. Does Villain's line make sense for value? No, it seems it doesn't. Hero reasons that such an active aggressive player is likely to bet most of his flushes on the turn for value as is normal practice for any Reg. Sure it's possible that Villain goes for a check raise on the turn with a flush, but it seems unlikely. Villain's river bet is so large that it really makes no sense ***for value*** with anything worse than a flush or perhaps **[A3 K3 43s]**. Villain shouldn't even have hands like **[AJ QQ-AA]** here having checked the turn anyway.

Overall, it just seems far too likely that this aggressive player is bluffing some amount much greater than the 36% that Hero needs him to be in order to call. This hand is an example of good hand reading skills put to practice. Hero deduces that though his hand is fairly weak in absolute terms, relative to Villain's range it's actually good very often due to how little sense his line makes for value.

Hero calls 36BB.

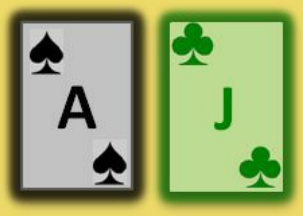
This kind of call is sometimes referred to as a 'hero call' as on the surface we're calling a huge bet with a weak hand. When you break down the problem, however, it seems more logical than heroic. I'm not even providing a definition box for this term because it's misleading and implies that there's something extraordinary about this call. In reality, these calls are actually less heroic than a call with a stronger hand that's closer to 0EV. This one isn't close as Villain is likely to be so unbalanced towards bluffs. I call it a 'bog standard call'.

Pre-flop End of Action Spots

End of Action Spots don't just occur on the river. Here's a relatively common pre-flop situation where we need to use a similar approach.

Hand 53

UTG – Aggro Fish (59/24) [37BB]
HJ – Unknown
CO – Unknown
BU – Unknown
SB – Passive Fish
BB – Hero

Dealt to Hero: 

UTG raises to 37BB (all in), HJ folds, CO folds, BU folds, SB folds, **Hero?**

This spot occurs from time to time in cash games though far less often than in tournament play. A Fish decides to jam his whole stack pre-flop like he were short stacked in the late stages of a tournament. The two-part process remains the same for Hero. 'How much equity do I need?' he asks and quickly realises that this all in raise size is very close to shoving 36BB into an empty pot. Of course the pot is not technically empty, but the 1.5BB of dead money makes little difference. Hero concludes that he needs ever so slightly less than 50% equity to call.

The true mathematical figure is 36 / (36 + 37.5) = **49%**.

It's also worth noting that the rake, which I typically ignore in the examples in this book, will eat up some of Hero's profits when he wins and so he can slightly increase this required equity up to 50%.

Now for part two of the thought process: Does Hero have 50% equity or more?

This can be tough to estimate in game unless you're familiar with how much equity hands have against certain ranges. This is why an important part of your active study time should be to play around with a poker calculator and get a rough feel for pre-flop equities, specifically for these all-in situations.

How wide should Villain be expected to shove here? Much will depend on recent history. If he's just lost a big pot and is launching death threats into the chat box, this Fish may be shoving any two cards.

If there's no reason to think he's tilted, his range will be a lot tighter.

If Villain is severely tilted, we can conclude that Hero should call assuming Villain's range to be absurdly wide. If he isn't severely tilted, a range around the vicinity of most pairs and the best broadways seems appropriate for this aggressive Fish.

Against **[66+ AJo+ KQo+ A9s+ KTs+ QJs+],** which is shown in grid format below, Hero has just **43%** equity and should fold. This way of describing ranges is a shorthand way of representing the range grids we've been using in the manual so far. We start by stating which pairs are in the range, then we move onto the offsuit hands, going from ace-high hands to king-high hands and so on, and then finally we state the suited hands in the range. See the figure below for a demonstration of how this translates into usual grid format.

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Villain Doesn't Shove Villain Shoves

Figure 40 - An Estimated Shoving Range

Hero folds unless Villain is tilted.

Final Thoughts

End of Action Spots can be tricky. Keep a clear head, take your time and logically work your way through the two-part thought process. You will miss things and make errors at first, but learn from those and you'll boost your pattern recognition skills significantly. This is an area of the game where improving the coherence of your thought process and picking up on crucial bits of information you'd once have missed will pay dividends. These spots also feel particular gross to play and this can hurt the mental game and induce tilt. Having a clear understanding of how to solve these situations will therefore stabilise your mental game to some extent and help keep confidence high.